

CS & IT ENGINEERING



Computer Network

Flow Control

Lecture No. - 07



By - Abhishek Sir



Recap of Previous Lecture



Topic

Sliding Window Protocol

Topic

Go Back N ARQ



Topics to be Covered



Topic

Go Back N ARQ

Topic

Selective Repeat ARQ

ABOUT ME



Hello, I'm **Abhishek**

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Statements for Linked Answer Questions:

Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts.
The propagation time is 25ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link).

#Q. What is the minimum number of bits (l) that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.

[GATE 2009]

A $l = 2$

B $l = 3$

C $l = 4$

D $l = 5$



Ans: D

Solution:-

$$\text{Packet Size} = 1000 \text{ bits}$$

$$\text{Bandwidth} = 10^6 \text{ bits / sec}$$

$$t_x = \frac{\text{Packet Size}}{\text{Bandwidth}} = \frac{10^3 \text{ bits}}{10^6 \text{ bits / sec}} = 1 \text{ ms}$$

$$t_p = 25 \text{ ms}$$

Minimum number of frames required to maximally pack them in transit

$$= \left\lceil \frac{t_p}{t_x} \right\rceil = \left\lceil \frac{25 \text{ ms}}{1 \text{ ms}} \right\rceil = 25$$

Transmitter's transmitting window size $(N) \geq 25$ ✓

For Sliding Window ARQ :

Total number of sequences = N

Minimum number of bits required for sequence number field (1)

$$= \lceil \log_2 [\text{Total number of sequences}] \rceil$$

$$= \lceil \log_2 (25) \rceil = 5 \text{ bits}$$

#Q. Suppose that the sliding window protocol is used with the sender window size of 2^l where l is the number of bits identified in the previous question and acknowledgments are always piggybacked. After sending 2^l frames, what is the minimum time the sender will have to wait before starting transmission of the next frame? (Identify the closest choice ignoring the frame processing time.)

[GATE 2009]

A 16 ms

B 18 ms

C 20 ms

D 22 ms

Ans: C

Solution:-

Piggybacking :

→ ACK always present inside packet header

→ Packet header contains :

1. Sequence Number (k bits)
2. ACK Number (k bits)

$$l_{x_A} = l_x$$

$$\text{Sender Window Size} = 2^l = 2^5 = 32$$

$$\begin{aligned} \text{Cycle Time} &= (t_x + t_p) + (t_{xA} + t_p) \\ &= (1 + 25) + (1 + 25) \text{ ms} \\ &= 52 \text{ ms} \end{aligned}$$

$$\begin{aligned} \text{minimum time the sender will have to wait} &= [\text{Cycle Time} - 2^l * t_x] \\ &= [52 \text{ ms} - (2^5) * 1 \text{ ms}] \\ &= (52 - 32) \text{ ms} \\ &= 20 \text{ ms} \end{aligned}$$



Topic : Go Back N ARQ



- Transmitter's transmitting window size = N ($N > 1$)
- Receiver's receiving window size = 1
- Total number of sequences = $N+1$ $[0 \text{ to } N]$

Total number of sequences =
Transmitter's transmitting window size
+ Receiver's receiving window size

$$\text{Sequence number} \leftarrow (\text{Frame number}) \bmod (N+1)$$



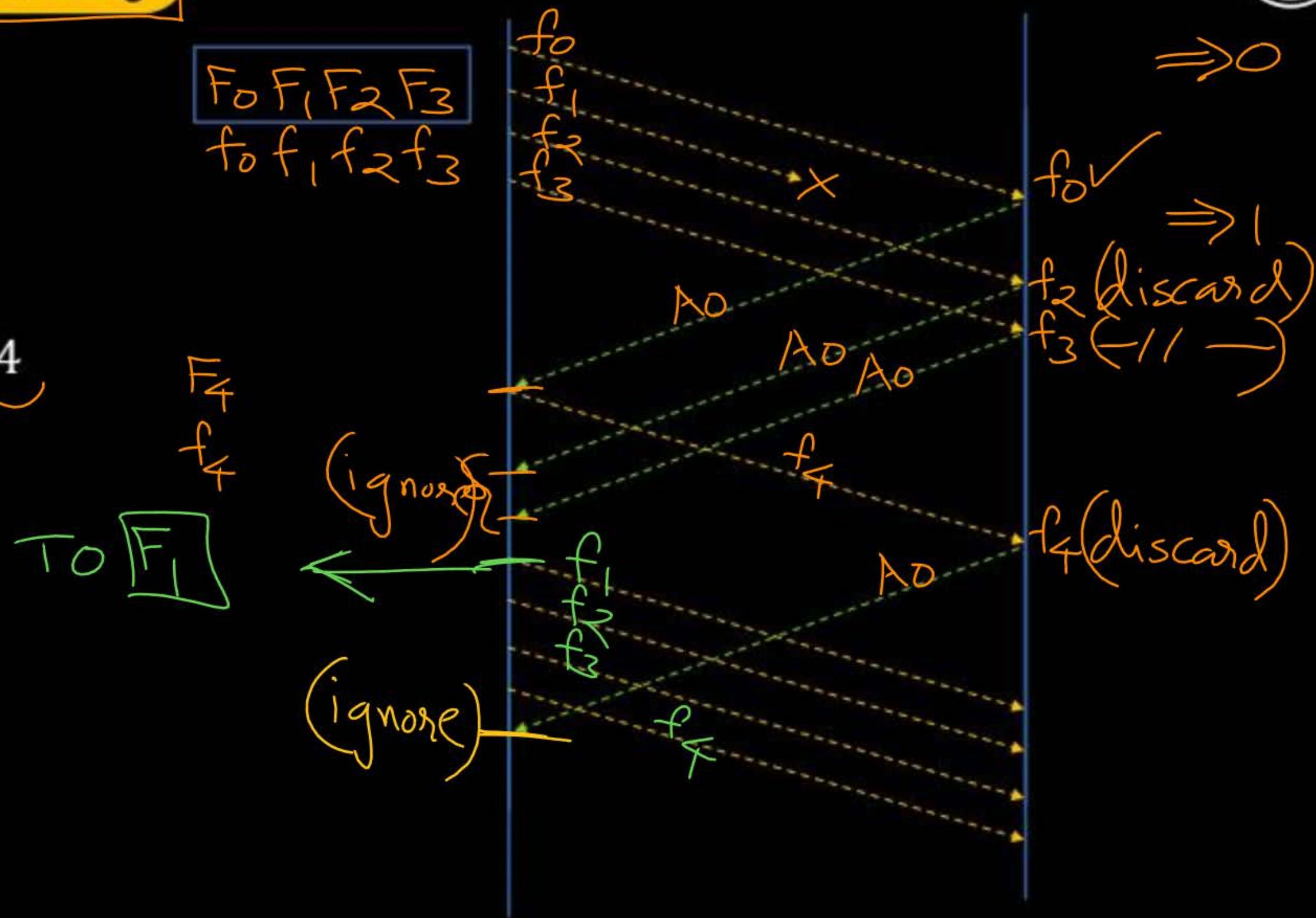
Topic : Go Back N ARQ



CASE V:

Go back 4 ARQ

Sequence Number = 0 to 4





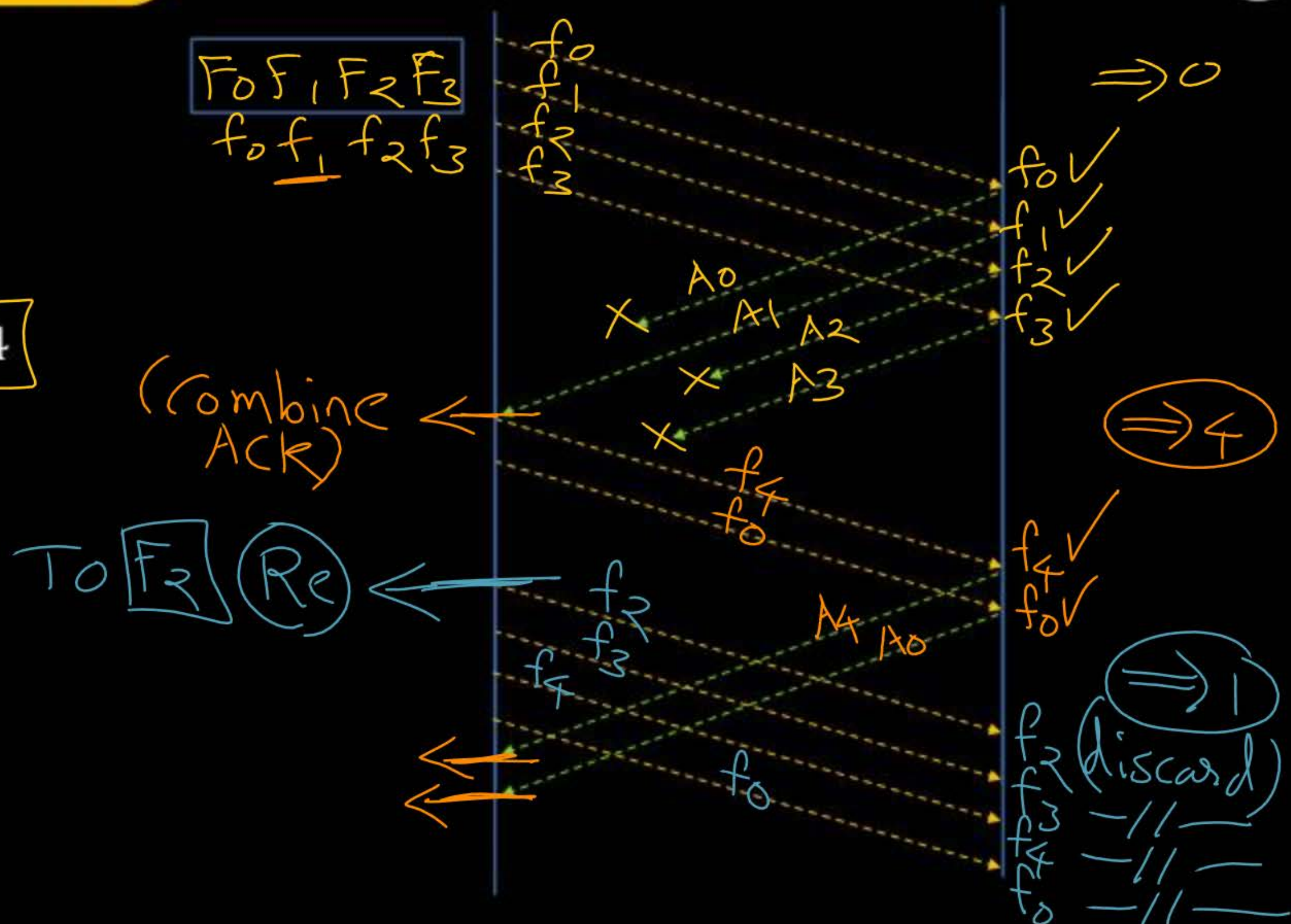
Topic : Go Back N ARQ



CASE VI :

Go back 4 ARQ

Sequence Number = 0 to 4





Topic : Go Back N ARQ



- Transmitter transmit **N** frames **without any acknowledgment**
- Receiver transmit “**individual acknowledgment**”
[for every successfully received frame]
- “**Cumulative (combine) acknowledgment**” may exist.
[Acknowledges more than one frame]



Topic : Go Back N ARQ



- Whenever transmitter gets time-out or received NACK, it retransmit all N frames [those resides in transmitting window]
- Receiver discard the frame which is out of order, and send ACK of the frame which is correctly received recently

Example 8 :-

Host A wants to send a file into 7 packets to ~~Station~~ ^{Host} B using go-back-n (window size 3) flow control strategy. If every 4th packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending the file to B?



Ans = 15



Transmitter

Receiver

Transmitter

Receiver

$F_1 F_2 F_3$

F_1
 F_2
 F_3

$F_1 \checkmark$
 $F_2 \checkmark$
 $F_3 \checkmark$

A_1
 A_2
 A_3

$F_4 F_5 F_6$

F_6
 F_5
 F_4

F_5
 F_6
 F_7 (discard)

A_3
 A_3

(ignore)

$T O F_4$

F_4
 F_5
 F_6

$F_4 \checkmark$
 F_6 (discard)

A_4
 A_4

$F_5 F_6 F_7$

(ignore)

$T O F_5$

F_5
 F_6
 F_7

F_7 (discard)

A_4

(ignore)

$F_5 \checkmark$

F_7 (discard)

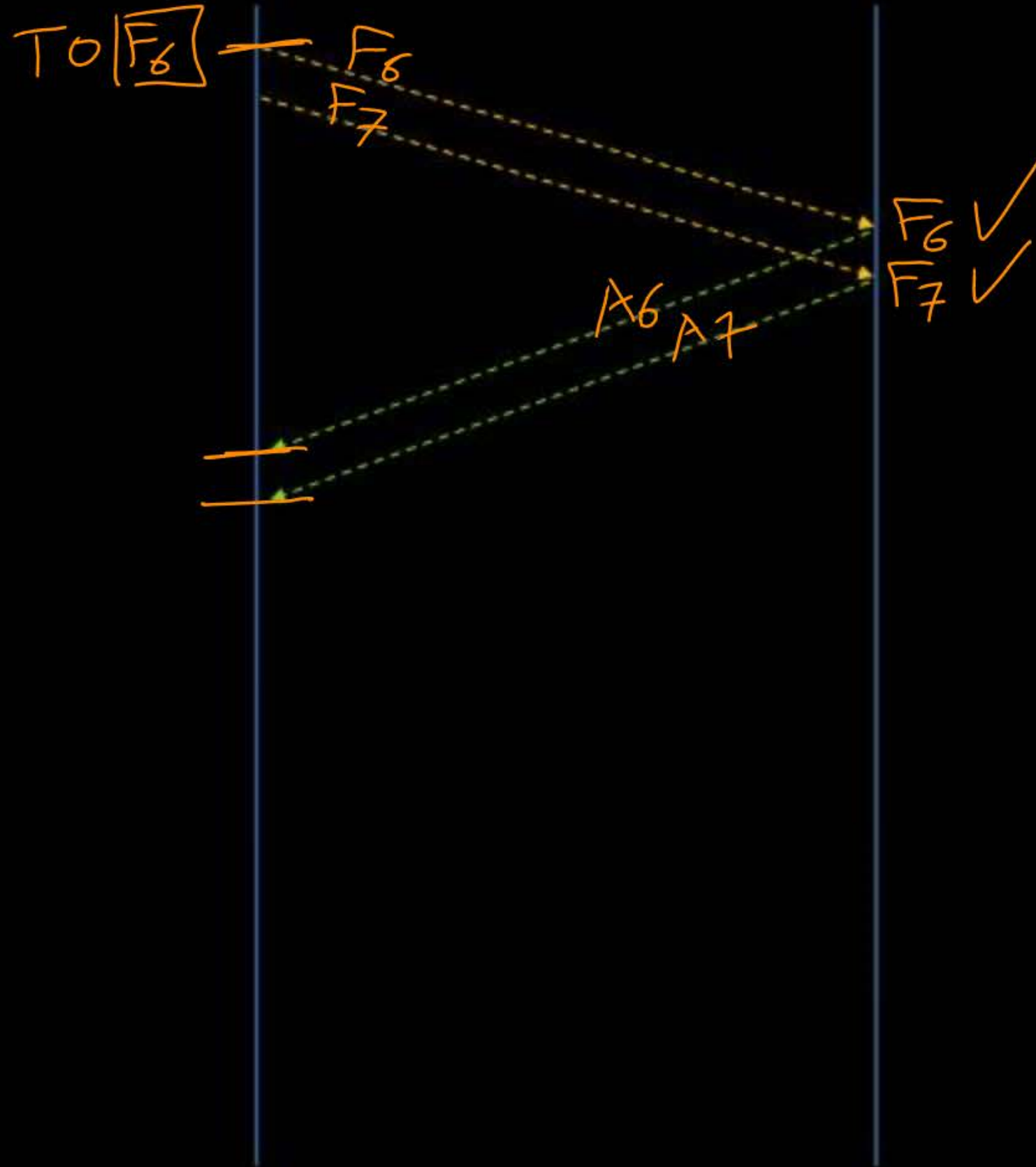
A_5

A_5

ignore

Transmitter

Receiver



Ans = 15

#Q. Station A needs to send a message consisting of 9 packets to Station B using a sliding window (window size 3) and go-back-n flow control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending the message to B?

A 12

B 14

C 16

D 18

[GATE 2006]

IIT-KGP
(H.W.)

#Q. Consider a network connecting two systems located 8000 kilometers apart. The bandwidth of the network is 500×10^6 bits per second. The propagation speed of the media is 4×10^6 meters per second. It is needed to design a Go-Back-N sliding window protocol for this network. The average packet size is 10^7 bits. The network is to be used to its full capacity. Assume that processing delays at nodes are negligible. Then, the minimum size in bits of the sequence number field has to be _____.

[GATE 2015, Set-3, 2-Mark]

IIT-K, H.W.

#Q. A 20 Kbps satellite link has a propagation delay of 400 ms. The transmitter employs the "go back n ARQ" scheme with n set to 10. Assuming that each frame is 100 bytes long, what is the maximum data rate possible?

[GATE 2004]

IIT-D, H.W.

- (A) 5Kbps
- (B) 10Kbps
- (C) 15Kbps
- (D) 20Kbps

#Q. A 1Mbps satellite link connects two ground stations. The altitude of the satellite is 36,504 km and speed of the signal is 3×10^8 m/s. What should be the packet size for a channel utilization of 25% for a satellite link using go-back-127 sliding window protocol? Assume that the acknowledgment packets are negligible in size and that there are no errors during communication.

[GATE 2008]

ISC, H.W.

- (A) 120 bytes
- (B) 60 bytes
- (C) 240 bytes
- (D) 90 bytes



Topic : Selective Repeat ARQ

→ Transmitter's transmitting window size = N ($N > 1$)

→ Receiver's receiving window size = N

→ Total number of sequences = $2N$

$[0 \text{ to } (2N-1)]$
→ $0 \text{ to } (N-1)$
→ $N \text{ to } (2N-1)$

Total number of sequences =

Transmitter's transmitting window size
+ Receiver's receiving window size

Sequence number $\leftarrow$ (Frame number) mod $(2N)$



2 mins Summary



Topic

Go Back N ARQ ✓

Topic

Selective Repeat ARQ



THANK - YOU

